

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCSIT
Course: Digital Systems

Semester: Fall

Year: 2024
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2=20]

1. Mention the basic difference between analog and digital system.
2. Mention the steps to get the Dual of any Boolean expression with example.
3. Write down the excitation table and characteristics equation of T flip-flop.
4. Find base X if $(211)_X = (152)_8$
5. What is an accumulator?
6. Explain K-map along with its importance.
7. How many flip-flops are there in 5 bits serial out shift registers?
8. Draw the block diagram of a combinational and sequential circuit.
9. Draw the diagram for 3-bit Johnson counter.
10. Define CMOS and mention its application.

Section "B"

Descriptive Answer Questions

Attempt **any five** questions. [5×10=50]

11. Simplify the following expression.

$$F = \sum_m (0, 2, 4, 6, 11, 15) + \sum_d (8, 10, 12, 13, 14)$$

Using k-map and draw the logic diagram using

- i. Basic gates only
- ii. NAND gates only

12. Differentiate full adder and half adder. Design full adder using 2 half adder and OR gate.
13. Convert the following number from the given base to the other bases indicated.
 - a. $(1156)_8 = ()_{16}$
 - b. $(3FD.ED)_{16} = ()_{10}$
 - c. $(206)_{10} = ()_5$
 - d. $(1011.1011)_2 = ()_{10}$
 - e. $(110011)_{\text{Gray code}} = ()_{\text{BCD}}$

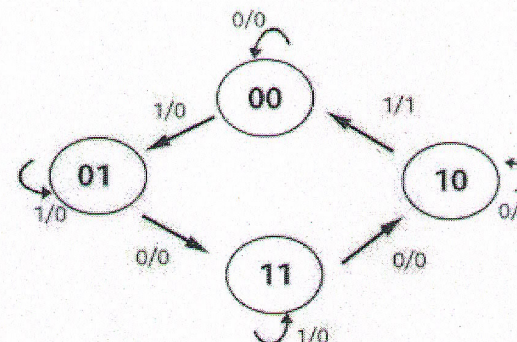
14. a. Describe the SOP & POS with examples.
b. What do you mean by divide by N Counter? Give example. Also give the examples of Synchronous and Asynchronous counter.
15. Explain in detail about the operation of SR flip-flop along with truth table, characteristics equation and circuit diagram.
16. Explain the operation of Serial IN Serial OUT (SISO) Shift Register along with its timing diagram.

Group "C"

Long Answer Questions

Attempt **any two** questions. [2×15=30]

17. a. Define ripple counter. Design a 3-bit Synchronous Counter.
b. Design a 4-bit ALU and explain its expression.
18. Draw the excitation table for RS, JK, T and D flip-flop. Design a sequential machine for the state diagram shown in figure below.



19. a. Design 1*8 demultiplexer using 1*2 demultiplexer.
b. Design 4-bit ALU.